Chapter 1 Differentiable Manifolds

1.1 Differerentiable Manifolds

1.1.1 Pseudo Group Axioms

Definition 1.1.1.1 A pseudogroup of transformations on a topological space S is a set Γ of transformations satisfying the following axioms:

- Each $f \in \Gamma$ is a homeomorphism of an open set of S onto another open set of S.

$$f_{\cong} : U \to V; U, V \in Open(S)$$

- If $f \in \Gamma$, then the restriction of f to an arbitrary open subset of domain of f is in Γ.(And it is still an homeomorphism)

$$f \in \Gamma \Rightarrow f|U \in \Gamma$$

- Let $U = \cup U_i$. Then,

$$f|U_i \in \Gamma \Rightarrow f \in \Gamma$$

- For every open U,

$$id_U \in \Gamma$$

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$$f \in \Gamma \Rightarrow f^{-1} \in \Gamma$$

- $f \in \Gamma$ is a homeomorphism of U onto V, $f' \in \Gamma$ is a homeomorphism of U' onto V', then

$$\Gamma \ni f' \circ f : f^{-1}(V \cap U') \to f'(V \cap U')$$

♣ **Example 1.1.1.2** (Pseudogroup $\Gamma^r(\mathbb{R}^n)$) Define:

$$\Gamma^r(\mathbb{R}^n) := \{f \in C^r(U, V) | U, V \in Open(\mathbb{R}^N)\}$$

There is an inclusion

$$\Gamma^{r+1}(\mathbb{R}^n) \hookrightarrow \Gamma^r(\mathbb{R}^n)$$

♣ **Example 1.1.1.3** Furthermore, Psudogroup

$$\mathcal{O}(\Omega)$$

1.1.2 Atlas

Definition 1.1.2.1 An **atlas** of a topological space M is a family of **charts** (U_i, φ_i) which satisfies:

- Each U_i is an open set of M, and

$$\cup U_i = M$$

- $\varphi_i : U_i \to S$ is a homeomorphism onto its image;
- If $U_i \cap U_j \neq \emptyset$, then

$$\varphi_j \circ \varphi_i^{-1} : \varphi_i(U_i \cap U_j) \to \varphi_j(U_i \cap U_j)$$

is an element of Γ.

Definition 1.1.2.2 An **compelete atlas** of M compatible with Γ is an atlas of M compatible with Γ.

Then, every atlas is contained in a unique compelete atlas.
Remark Suppose A, A' be two atlases. Then, if they are **compatible**

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1.1.3 Differentiable Manifold

Definition 1.1.3.1 ...

1.1.4 Differentiable Mapping

1.1.5 Differentiable Curve, Tangewnt Vectors

Differentiable Curve

Definition 1.1.5.1 ...

Tangent Vectors

Definition 1.1.5.2 Let $\mathfrak{F}(p), p \in M$ be the equivalence class of differentiable functions at p,

$$\mathfrak{F}(p) := \{(U, f) | f \in C^1(U, \mathbb{R}), p \in U \subseteq X \text{ open } \} / \sim \qquad (U, f) \sim (V, g) \Leftrightarrow f|U \cap V = g|U \cap V$$

$\mathfrak{p}$ is an $\mathbb{R}$-linear space.

Then, let $x : [a, b] \to M$ be a C^1-differentiable function where

$$x(t_0) = p$$

Then, define

$$X : \mathfrak{F}(p) \to \mathbb{R}$$
$$[(U, f)] \mapsto (\frac{df(x(t))}{dt})|_{t_0}$$

In other word, Xf is derivative of f in the direction of the curve $x(t)$ at $t = t_0$.
The function X satisfies the following properties:

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Lie Bracket

1.2 Tensor Algebra, Tensor Fields

Given in Smooth manifold.

1.3 Lie Groups

Given in Lie theory.

1.4 Fibre Bundles

1.4.1 Differentiable Principle Fibre Bundle(G-Bundles)

Definition 1.4.1.1 A **differentiable principle fibre bundle** over M with group G consists of $P(M, G, \pi)$, where:

- G acts freely on P on the right:

$$P \times G \to P \qquad (u, a) \mapsto ua = R_a u \in P$$

- M is quotient space of P, and the canonical projection

$$\pi : P \to M$$

is differentiable;

- P is locally trivial:

$$\pi^{-1}(U) \cong U \times G$$

in the sense of diffeomorphism $\psi : \pi^{-1}(U) \to U \times G, u \mapsto (\pi(u), \varphi(u)); \varphi(ua) = (\varphi(u))a, u \in \pi^{-1}(U), a \in G$.

Take the following notion of fibre bundles $P(M, G, \pi)$.

Example: Lie Group

Fibre of Differentiable Principle Fibre Bundle

1.4.2 Fundamental Vector Field

Definition 1.4.2.1 Given a principle fiber bundle $P(M, G, \pi)$, the action of G on P induces a homomorphism σ

of the Lie algebra $\mathfrak{g}$ of G into the Lie algebra $\mathfrak{X}(P)$ of vector fields on P.

$$\mathfrak{g} = L(G) \to \mathfrak{X}(P)$$

Constructions:

- Exponential Maps:
- Consider the action

$$\Phi_u : G \to P \quad g \mapsto u \cdot g$$

Then,

$$\sigma : \mathfrak{g} = T_e G \to \mathfrak{X}(P) \sigma(A)_u = (d\Phi_u)_e(A)$$

Take the following notation

$$A^* := \sigma A$$

Furthermore, we have

♠ **Proposition 1.4.2.2**

$$\mathfrak{g} \to T_u P \qquad A \mapsto (A^*)_u$$

is linear isomorphism.

∘ **Proof**

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□

♠ **Proposition 1.4.2.3** Let A^* be fundmanetal vector field, then there exists correspondence

$$A \in \mathfrak{g} \Rightarrow A^* \in \mathfrak{X}(P)$$
$$(ad(a^{-1}))A \in \mathfrak{g} \Rightarrow (R_a)_* A^*$$

∘ **Proof** ... □

The mapping σ is injective, but not surjective. But we could define a Lie-subalgebra of $\mathfrak{X}(P)$, so that $\sigma : L(G) \to \mathfrak{X}(P)$ is bijective.

Definition 1.4.2.4 ...

♠ **Proposition 1.4.2.5** Let M be a manifold. $\{U_\alpha\}$ is an open covering of M and G a Lie group.
Given mapping

∘ **Proof** ... □

Chapter 2 Theory of Connections

2.1 Connections in a Principle Vector Bundle

Remark Firstsly, the fibre through u is

$$E_{\pi(u)}$$

Definition 2.1.0.1 Let (P, M, G, π) be a principle fibre bundle over a manifold M. For each $u \in P$, $T_u(P)$ be the tangent space of P at u, and G_u be the subspace of $T_u(P)$ consisting of vectors tangent to fibre through u.

$$G_u := T_u(E_{\pi(u)}) = Ker(\pi_u)$$

- $T_u(P) = G_u \oplus Q_u$; G_u is the verticle subspace, Q_u is the horizontal subspace;
  Take the following notation vX, hX.
- $Q_{ua} = (R_a)^* Q_u; u \in P, a \in G$
- Q depends differentiably on u;

A **connection** is a distribution $u \mapsto Q_u$.

Remark 1 The tangent space is the formel tangent space, not rely on the open subset and the tangent line via limit.

2.1.1 Connection Form

◇ **Lemma 2.1.1.1** $\sigma_u : \mathfrak{g} \to T_u(P)$ is injective, $u \in P$,with induced linear isomorphism

$$\sigma_u : \mathfrak{g} \to V_u(P) \qquad A \mapsto (A^*)_u$$

∘ **Proof** ... □

Definition 2.1.1.2 Given a connection Γ in P. We define the 1-form P with the value in Lie algebra of G as follows: $\omega \in \Omega^1(P, \mathfrak{g})$,

$$\omega_p : T_u(P) \to \mathfrak{g} \qquad X \mapsto A$$

such that $(A^*)_u = vX$.

Remark 1 $\mathfrak{g} \to G_u, A \mapsto (A^*)_u$ is a linear isomorphism.

♠ **Proposition 2.1.1.3** The connection form ω of a connection satisfies the following conditions:

- $\omega(A^*) = A$;
- $(R_a)^* = ad(a^{-1})\omega$, that is

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for every $a \in G$ and every vector field X on P.

∘ **Proof** ... □

♠ **Proposition 2.1.1.4** ...

∘ **Proof** ... □

♠ **Proposition 2.1.1.5** ...

2.2 Existence and Extension of Connections

▶ ***Theorem*** ...

∘ **Proof** ... □

2.3 Parallelism

2.3.1 Horizontal Lift

Definition 2.3.1.1 Let $\tau = x(t)$ be a piecewise differentiable curve of C^1 in M. A **horizontal lift** or **lift** is a **horizontal curve**

$$\tau^* = u(t)$$

such that $x(t) = \pi(u(t))$.
Horizontal: Piecewise differentiable, the tangent vectors are horizontal.

♠ **Proposition 2.3.1.2** (Existence, Uniqueness)

∘ **Proof** ... □

2.3.2 Parallel Displacement

2.4 Holonomy Groups

Another Reference: Riemannian Holonomy Groups and Calibrated Geometry by Dominic D. Joyce

Definition 2.4.0.1 For point $x \in M$, denote

2.5 Curvature Form, and Structure Equation

Definition 2.5.0.1 Let $P(M, G, \pi)$ be a PFB and representation on G on V, with $\rho(a) \in End(V), a \in G$. $\rho(ab) = \rho(a)\rho(b)$.

A **pseudotensorial form** of degree on P of type (ρ, V) is a V-valued r-form φ such that

$$R_a^*\varphi = \rho(a^{-1}) \cdot \varphi; a \in G$$

Such a form is called **tensorial form** if it is horizontal in the sense of

$$\varphi(X_1, \dots, X_i, \dots, X_r) = 0; X_i \in V_i P$$

♠ **Proposition 2.5.0.2**

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∘ **Proof** ... □

Especially:

Definition 2.5.0.3

- The connection form ω is a pseudotensorial 1-form of type $ad(G)$;
- $D\omega$ is called **tensorial 2-form** of type $ad(G)$, denoted by Ω, is called the **curvature form** of ω.

2.5.1 Structure Equation

▶ ***Theorem*** (Structure Equation,E.Cartan) Let ω be a connection form and Ω be its curvature form, then

$$\Omega(X, Y) = d\omega(X, Y) + \frac{1}{2}[\omega(X), \omega(Y)]$$

∘ **Proof** ... □

♠ **Proposition 2.5.1.1** ...

∘ **Proof** ... □

2.5.2 Bianchi's Identity

▶ ***Theorem*** (Bianchi)

$$D\Omega = 0$$

∘ **Proof** ... □

2.6 Mappings of Connections

2.7 Reduction Theorem

2.8 Holonomy Theorem

2.9 Flat Connections

2.10 Local, and Infinitesimal Holonomy Groups

2.11 Invariant Connections

Chapter 3 Linear Affine Connections

3.1 Connections in Vector Bundles

3.1.1 Fibre Metrics

3.2 Linear Connections

Chapter 4 Curvature and Space Forms

Chapter 5 Transformations

Chapter 6 Submanifolds

Chapter 7 Variation of Length Integral

Chapter 8 Homogeneous Spaces

Chapter 9 Symmetric Spaces